Calculus I Lesson 40

1) $\int \frac{15}{x} \cdot dx = \underline{\quad ? \quad}$

Hint: $\frac{15}{x} = 15 \cdot \frac{1}{x}$

Hint: $\int \frac{1}{x} dx = \ln|x| + C \qquad \int \frac{1}{x \pm b} dx = \ln|x \pm b| + C \qquad \int \frac{1}{ax \pm b} dx = \frac{1}{a} \ln|x \pm b| + C$

2) Find $\int \frac{1}{x+12}dx = \underline{\quad ? \quad}$

Hint: $\int \frac{1}{x}dx = \ln|x| + C \qquad \int \frac{1}{x \pm b}dx = \ln|x \pm b| + C \qquad \int \frac{1}{ax \pm b}dx = \frac{1}{a}\ln|x \pm b| + C$

3) Find $\int \frac{1}{4x+5}\,dx =$ ___?___

Hint: $\int \frac{1}{x}\,dx = \ln|x| + C \qquad \int \frac{1}{x \pm b}\,dx = \ln|x \pm b| + C \qquad \int \frac{1}{ax \pm b}\,dx = \frac{1}{a}\ln|x \pm b| + C$

4) Find $\int \frac{4x}{x^2-5}dx = 4\int \frac{1}{x^2-5}(x)\,dx.$

Hint: Let $u = x^2 - 3$

Hint: $\int \frac{1}{x}dx = \ln|x| + C$ $\quad \int \frac{1}{x \pm b}dx = \ln|x \pm b| + C$ $\quad \int \frac{1}{ax \pm b}dx = \frac{1}{a}\ln|x \pm b| + C$

a) $\frac{du}{dx} = \underline{\quad ? \quad}$ and $\frac{1}{2}du = \underline{\quad ? \quad}$

b) Write the indefinite integral $\int \frac{4x}{x^2-5} \cdot dx = 4 \cdot \int \frac{1}{x^2-5} \cdot x \cdot dx$ in terms of u and then evaluate.

$$4 \cdot \int \frac{1}{x^2-5} \cdot x \cdot dx = \int \left(\underline{\quad ? \quad}\right) \cdot du = \underline{\quad ? \quad}$$

c) $\int \frac{4x}{x^2-5} \cdot dx = \underline{\quad ? \quad}$ (Note: Answer must be in terms of x.)

5) Find $\int \frac{4x^3-1}{x^4-x}dx = \int \frac{1}{x^4-3x}\left(4x^3-1\right)dx.$

Hint: Let $u = x^4 - x$

Hint: $\int \frac{1}{x}dx = \ln|x| + \mathrm{C}$ $\quad \int \frac{1}{x \pm b}dx = \ln|x \pm b| + \mathrm{C}$ $\quad \int \frac{1}{ax \pm b}dx = \frac{1}{a}\ln|x \pm b| + \mathrm{C}$

a) $\frac{du}{dx} = \underline{\quad ? \quad}$ and $du = \underline{\quad ? \quad}$

b) Write the indefinite integral $\int \frac{4x^3-1}{x^4-x} \cdot dx = \int \frac{1}{x^4-x} \cdot \left(4x^3-1\right) \cdot dx$ in terms of u and then evaluate.

$$\int \frac{1}{x^4-x} \cdot \left(4x^3-1\right) \cdot dx = \int \left(\underline{\quad ? \quad}\right) \cdot du = \underline{\quad ? \quad}$$

c) $\int \frac{4x^3-1}{x^4-x} \cdot dx = \underline{\quad ? \quad}$ (Note: Answer must be in terms of x.)

6) Find $\int \frac{x^2-2}{x}dx = \int\left(\frac{x^2}{x}-\frac{2}{x}\right)dx = \int\left(x-\frac{2}{x}\right)dx = \int(x)\,dx - \int\left(\frac{2}{x}\right)dx$

Hint: $\int \frac{1}{x}dx = \ln|x| + C \qquad \int \frac{1}{x\pm b}dx = \ln|x\pm b| + C \qquad \int \frac{1}{ax\pm b}dx = \frac{1}{a}\ln|x\pm b| + C$

a) $\int(x)\,dx =$ ___?___

b) $\int\left(\frac{2}{x}\right)dx = 2\cdot\int\frac{1}{x}dx =$ ___?___

c) $\int \frac{x^2-2}{x}dx =$ ___?___

7) Find $\int \frac{x^2-5x+12}{x+1}\,dx$

Hint: Use Long Division to divide $\frac{x^2-5x+12}{x+1}$ first.

Hint: $\int \frac{1}{x}\,dx = \ln|x| + C \qquad \int \frac{1}{x \pm b}\,dx = \ln|x \pm b| + C \qquad \int \frac{1}{ax \pm b}\,dx = \frac{1}{a}\ln|x \pm b| + C$

a) Divide $\frac{x^2-5x+12}{x+1} = \underline{\quad ? \quad}$

b) $\int \frac{x^2-5x+12}{x+1}\,dx = \underline{\quad ? \quad}$

8) Note: $\int \frac{(\ln x)^2}{x} dx = \int (\ln x)^2 \cdot \frac{1}{x} dx$

Hint: Let $u = \ln x$.

Hint: $\int \frac{1}{x} dx = \ln|x| + \text{C}$ $\qquad \int \frac{1}{x \pm b} dx = \ln|x \pm b| + \text{C}$ $\qquad \int \frac{1}{ax \pm b} dx = \frac{1}{a} \ln|x \pm b| + \text{C}$

a) $\frac{du}{dx} =$ ___?___

b) $du =$ ___?___

c) $\int \frac{(\ln x)^2}{x} dx =$ ___?___